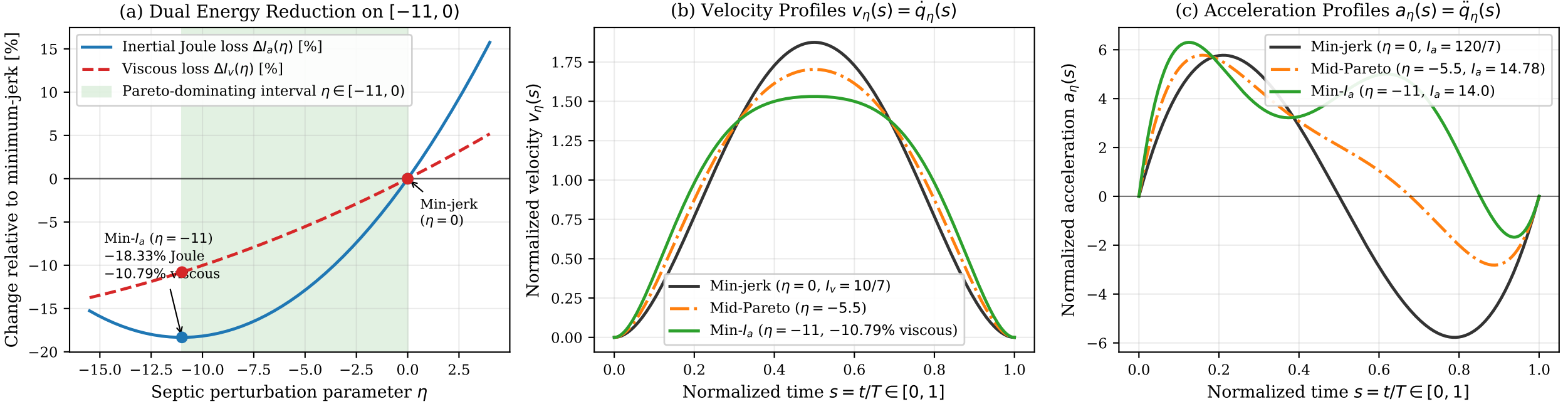
(a) Dual Energy Reduction on $[-11, 0)$
Inertial Joule loss $\Delta I_a(\eta)$ [%]
Viscous loss $\Delta I_v(\eta)$ [%]
Pareto-dominating interval $\eta \in [-11, 0)$
Min-jerk
($\eta = 0$)
Min-I_a ($\eta = -11$)
−18.33% Joule
−10.79% viscous
Change relative to minimum-jerk [%]
Septic perturbation parameter η
(b) Velocity Profiles $v_\eta(s) = \dot{q}_\eta(s)$
Min-jerk ($\eta = 0$, $I_v = 10/7$)
Mid-Pareto ($\eta = -5.5$)
Min-I_a ($\eta = -11$, −10.79% viscous)
Normalized velocity $v_\eta(s)$
Normalized time $s = t/T \in [0, 1]$
(c) Acceleration Profiles $a_\eta(s) = \ddot{q}_\eta(s)$
Min-jerk ($\eta = 0$, $I_a = 120/7$)
Mid-Pareto ($\eta = -5.5$, $I_a = 14.78$)
Min-I_a ($\eta = -11$, $I_a = 14.0$)
Normalized acceleration $a_\eta(s)$
Normalized time $s = t/T \in [0, 1]$